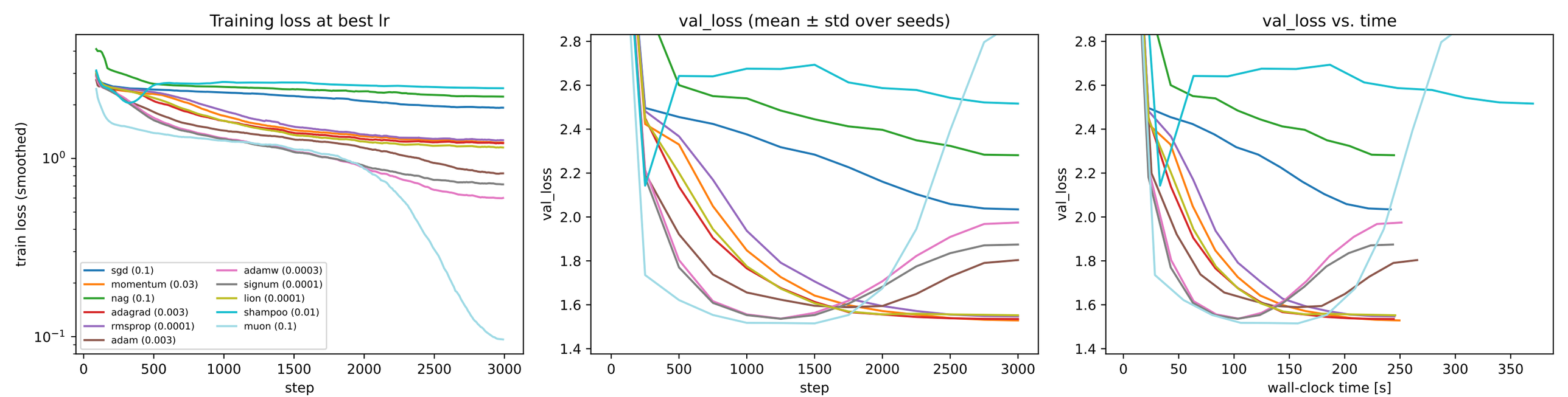
Training loss at best lr
train loss (smoothed)
10^0
10^{-1}
step
sgd (0.1)
momentum (0.03)
nag (0.1)
adagrad (0.003)
rmsprop (0.0001)
adam (0.003)
adamw (0.0003)
signum (0.0001)
lion (0.0001)
shampoo (0.01)
muon (0.1)
val_loss (mean ± std over seeds)
val_loss
step
val_loss vs. time
val_loss
wall-clock time [s]